FOR EDITH · OBSTETRICS & GYNAECOLOGY JUNIOR CLERKSHIP

Malignant Lesions of the Cervix

Tutorial | Week 5 · Bleeding, cancer & fertility | Gynaecology | Dr Kaduyu

Cervical cancer is a preventable, HPV-driven cancer that is the commonest cancer and cancer killer of Ugandan women; prevention rests on HPV vaccination and screen-and-treat of pre-cancer, and treatment of invasive disease depends on FIGO 2018 stage — surgery for early disease, chemoradiation at the Uganda Cancer Institute for locally advanced disease, and palliative care for the many who present late.

Must know before the ward round

1. Persistent high-risk HPV (types 16 and 18 cause about 70%) is necessary for cervical cancer; E6 degrades p53 and E7 inactivates Rb, releasing the brakes on the cell cycle.
2. Women living with HIV are about 6 times more likely to develop cervical cancer — WHO recommends screening them from age 25, every 3–5 years; the general population from 30, every 5–10 years with HPV DNA.
3. WHO 90-70-90 by 2030 — 90% of girls fully vaccinated by 15, 70% of women screened with a high-performance test by 35 and again by 45, 90% of women with pre-cancer or cancer treated; elimination threshold is under 4 per 100,000 women.
4. Screen-and-treat — VIA (3–5% acetic acid, read at 1 minute) or HPV test, then ablation (thermal ablation ≥100 °C for 20–30 s, or cryotherapy) if eligible; LEEP if the lesion is large (>75%), enters the canal or the SCJ is not fully visible; biopsy if cancer is suspected.
5. FIGO 2018 — IB1 <2 cm, IB2 2–<4 cm, IB3 ≥4 cm; IIB parametrium; IIIB pelvic wall or hydronephrosis; IIIC1 pelvic nodes, IIIC2 para-aortic nodes; IVA bladder/rectal mucosa; IVB distant.
6. Early disease (IA2–IB2, IIA1) → radical hysterectomy + pelvic lymphadenectomy (or chemoradiation); locally advanced (IB3–IVA) → concurrent chemoradiation — EBRT 45–50 Gy in 25 fractions + weekly cisplatin 40 mg/m² + brachytherapy, total time ≤56 days.
7. Over 80% of Ugandan women reach the Uganda Cancer Institute with advanced disease; the commonest cause of death is renal failure from ureteric obstruction.
8. Uganda gives a single dose of HPV vaccine to 10-year-old girls (since 2025); immunocompromised girls (e.g. HIV) still need at least 2 doses.
9. Symptomatic women are diagnosed, not screened — PCB, PMB or offensive discharge needs a speculum look and a punch biopsy from the edge of any growth; never ablate a lesion that looks invasive.
10. Clinical FIGO staging — EUA with speculum, bimanual and rectovaginal examination (parametria), cystoscopy/proctoscopy if bladder or rectum is suspected, renal ultrasound or IVU (hydronephrosis = IIIB whatever the tumour size) and a chest X-ray; bullous oedema alone is not IVA.
11. Minimum workup before referral to UCI — histology, Hb (keep ≥10 g/dL during radiotherapy), urea/creatinine, renal ultrasound, CXR, HIV status with CD4 and viral load.
12. Cisplatin is withheld if creatinine clearance is below about 50 mL/min — drain obstructed kidneys (nephrostomy) first; heavy tumour bleeding gets tranexamic acid 1 g IV, a vaginal pack with a urinary catheter and urgent haemostatic radiotherapy.

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1. Definition

Cervical cancer is a malignant tumour arising from the epithelium of the cervix, almost always in the **transformation zone**, caused by **persistent infection with high-risk human papillomavirus (HPV)**. It develops slowly (10–20 years) through **pre-cancerous stages (CIN/SIL)** that can be detected and treated cheaply. This makes it **the most preventable cancer**: **vaccinate girls, screen women and treat pre-cancer the same day**.

Yet most Ugandan women still present with **advanced disease** (offensive bloody discharge, pain, fistulae and renal failure) when cure is unlikely. Your job as a clinician is therefore threefold:

1. **Never miss an opportunity to screen** (every woman aged 25/30–49, and every woman living with HIV from 25).
2. **Recognise and stage invasive cancer** quickly, get a biopsy, and refer.
3. **Relieve suffering**: bleeding, pain, odour, fistula and renal failure through good **palliative care**.

> **RED FLAG**
> **Post-coital bleeding, post-menopausal bleeding, or offensive blood-stained discharge = cervical cancer until proven otherwise.** Look at the cervix and biopsy anything suspicious. A screening test is not enough for a symptomatic woman.

Term	Meaning
HPV	Double-stranded DNA virus; **high-risk types** (16, 18, 31, 33, 35, 45, 52, 58 …) cause cancer; low-risk types (6, 11) cause genital warts
CIN (cervical intraepithelial neoplasia)	Histological pre-cancer: **CIN 1** (lower third of epithelium; mostly a transient HPV infection), **CIN 2** (lower two-thirds), **CIN 3** (full thickness, includes carcinoma in situ)
SIL (squamous intraepithelial lesion, Bethesda/LAST terminology)	**LSIL** = CIN 1 (and HPV effect); **HSIL** = CIN 2 and CIN 3
AIS	Adenocarcinoma in situ: glandular pre-cancer, often in the canal, hard to see on VIA
Micro-invasive (stage IA)	Invasion seen only microscopically, depth <5 mm
Screen-and-treat	Treatment decided on the basis of a positive screening test alone (no colposcopy or histology)
Screen, triage and treat	Positive primary test (e.g. HPV) is followed by a second (triage) test (e.g. VIA, colposcopy, cytology) before treatment
Ablation	Destroys the TZ: **cryotherapy** (freezing) or **thermal ablation** (heat)
Excision	Removes the TZ for histology: **LEEP/LLETZ**, cold-knife cone

2. Epidemiology

- **Burden**: **cervical cancer is the commonest cancer and the leading cause of cancer death among women in Uganda**. GLOBOCAN 2022: age-standardised incidence about **53.8** and mortality about **40.6 per 100,000** women, more than 10 times the WHO elimination threshold (<4 per 100,000); one of the highest rates in the world. Over 4,000 Ugandan women die of it every year. It accounts for about a third of cancers in Ugandan women.
- **Late presentation**: about **80% of women reaching the Uganda Cancer Institute (UCI)** have advanced disease (stage IIB or worse). Only a small fraction of those prescribed chemoradiation complete it on time (one UCI study found **only 12% compliance** with the full treatment course), because of distance, cost, radiotherapy queues and machine downtime.
- **Screening coverage** remains very low overall (single-digit percentages in several analyses), concentrated at higher-level facilities and HIV clinics, against a national target of about 80% of women aged 30–49.
- **HIV**: Uganda has high HIV prevalence among women; HIV reduces clearance of HPV, increases persistence, multiple-type infection and faster progression. Women living with HIV have about **6 times** the risk. About a quarter of women with cervical cancer at UCI are known to be HIV-positive.
- **Services**: radiotherapy is concentrated at **UCI in Kampala** (with expansion to regional cancer centres such as Gulu and Mbarara for chemotherapy and, increasingly, radiotherapy). Surgery for early disease is done by gynaecological oncologists at Mulago, UCI and some large referral/private-not-for-profit hospitals (Mengo, Nsambya, Lubaga, Mbarara, Gulu).
- **Palliative care**: Uganda is a world pioneer: **oral morphine** is manufactured locally and distributed free through public facilities, and trained nurses and clinical officers can prescribe it.

Services by level of care Verify *compiled from the source notes, not an MoH service list*

Level	What is available
HC III	Counselling, VIA (where trained), HPV self-sample collection in some programmes, referral
HC IV / HIV clinic	Screen-and-treat (VIA, HPV sample, thermal ablation/cryotherapy), punch biopsy, HemoCue, basic ultrasound, rapid HIV test, oral morphine
General hospital	Biopsy, Hb, creatinine, renal ultrasound, CXR, HIV status (core staging set), transfusion, vaginal packing
Regional referral hospital / regional cancer centre	LEEP, colposcopy, histology, EUA with cystoscopy, IVU, CT at some, chemotherapy (Gulu, Mbarara), increasingly radiotherapy
UCI / Mulago	MDT, MRI/CT, radical hysterectomy, EBRT + brachytherapy, chemoradiation, nephrostomy, palliative care unit

3. Aetiology and Risk Factors

Cause: persistent infection with high-risk HPV is **necessary**. **HPV 16 and 18 cause about 70%** of cervical cancers. Think "**HPV exposure + failure to clear it**".

Risk factor	Why
Early sexual debut, multiple partners, partner with many partners	More HPV exposure while the TZ is immature
HIV / immunosuppression	Failure to clear HPV; faster progression
Smoking	Carcinogens concentrated in cervical mucus; local immunosuppression

Risk factor	Why
High parity, early first pregnancy	Prolonged ectropion/metaplasia during pregnancy; hormonal and trauma effects
Long-term COC use (>5 years)	Modest increase; reverses after stopping
Never screened	The single biggest risk factor for invasive cancer
Other STIs (chlamydia, HSV-2)	Co-factors of inflammation
Low socio-economic status	Less screening, later presentation

4. Pathophysiology

How HPV causes cancer

1. **Infection**: HPV enters **basal cells** of the transformation zone through micro-abrasions during sex. Most infections (about 90%) are **cleared within 1–2 years** by cell-mediated immunity.
2. **Persistence**: in a minority (more with HIV, smoking), high-risk HPV persists.
3. **Integration**: viral DNA integrates into the host genome, typically disrupting the viral **E2** gene. E2 normally represses E6 and E7, so its loss causes **over-expression of E6 and E7**.
4. **Oncoproteins**:
   - **E6** binds the cellular ubiquitin ligase E6AP and targets **p53 for degradation** → no apoptosis, no DNA-damage checkpoint → mutations accumulate. E6 also activates telomerase (immortality).
   - **E7** binds and inactivates **retinoblastoma protein (pRb)** → releases **E2F** → uncontrolled entry into S-phase (proliferation). A surrogate marker of this is **p16 over-expression** used in histology.
5. **Progression**: CIN 1 → CIN 2 → CIN 3 → invasion through the basement membrane (usually **10–20 years**: the window for screening). In women with HIV, progression can be faster.
6. **Local spread**: into the **parametria** (lateral) → **ureters** (hydronephrosis → renal failure), pelvic side wall (sciatic pain, leg oedema); into the **vagina** (down), the **bladder** (anterior) and **rectum** (posterior) → fistulae; into the **uterine body** (up).
7. **Lymphatic spread** (orderly): parametrial → obturator/internal and external iliac → common iliac → **para-aortic** → supraclavicular (Virchow) nodes.
8. **Haematogenous spread** (late): lungs, liver, bone.

> CLINICAL PEARL
> **Why screening works**: the long, detectable pre-invasive phase + a visible, accessible organ + cheap, effective ablation. **Why vaccination works**: the vaccine produces neutralising antibodies that block infection **before** exposure, so it must be given **before sexual debut** (hence 9–14-year-old girls; Uganda targets 10-year-olds).

Why HIV matters: HIV depletes CD4 cells and impairs cell-mediated immunity, so HPV persists, multiple types co-exist, lesions are larger, recur more after treatment, and progress faster. **That is why WHO screens women living with HIV earlier (25), more often (every 3–5 years), uses screen–triage–treat, and re-tests at 12 months after treatment.**

High-risk HPV infects basal cells of the transformation zone	E7
↓	
Most cleared in 1-2 years	Persistence: HIV, smoking
↓	
Integration: E2 lost, E6 and E7 over-expressed	
↓	
E	
↓	
CIN 1 to CIN 3 over 10-20 years	
↓	
Invasion through basement membrane	
↓	
Local spread: parametria, vagina, bladder, rectum	Lymphatic: pelvic then para-aortic nodes
↓	
Ureteric obstruction: renal failure, death	Haematogenous: lung, liver, bone

Feature links

- Fragile, neovascular tumour surface → post-coital, inter-menstrual, contact bleeding.
- Tumour erosion of vessels → heavy, sometimes torrential haemorrhage → anaemia.
- Tumour necrosis and anaerobic infection → offensive, watery, blood-stained discharge.
- Parametrial and side-wall spread → back pain, sciatic pain, leg oedema.
- Ureteric compression → hydronephrosis → oliguria, uraemia (vomiting, hiccups, itching, confusion, acidotic breathing).
- Bladder or rectal invasion (or radiation necrosis) → haematuria, rectal bleeding, tenesmus, VVF/RVF.
- Canal occluded by tumour → pyometra, fever.
- Pelvic node and vein involvement + hypercoagulability → lymphoedema, DVT/PE.
- Para-aortic → supraclavicular spread → left supraclavicular node.
- Haematogenous spread → cough (lung), jaundice/hepatomegaly (liver), bone pain.
- Hypoxic (anaemic) tumour → radioresistance → Hb kept ≥10 g/dL.

5. Classification / Staging

Histological types: **squamous cell carcinoma (about 75–80%)**, **adenocarcinoma (about 15–20%**, increasing; HPV 18-associated; can hide in the canal), adenosquamous, and rare neuroendocrine (small cell) tumours.

FIGO 2018 staging (with 2019 corrigendum)

Stage	Description
I	Confined to the cervix (extension to the uterine body disregarded)
IA	Invasive carcinoma diagnosed only by microscopy, maximum depth of invasion **<5 mm**
IA1	Stromal invasion **<3 mm** deep
IA2	Stromal invasion **≥3 mm and <5 mm**
IB	Invasion ≥5 mm (larger than IA), confined to the cervix

Stage	Description
IB1	Tumour **<2 cm** in greatest dimension
IB2	Tumour **≥2 cm and <4 cm**
IB3	Tumour **≥4 cm**
II	Beyond the uterus, but not to the lower third of the vagina or the pelvic wall
IIA	Upper two-thirds of vagina, **no parametrial** involvement (IIA1 <4 cm; IIA2 ≥4 cm)
IIB	**Parametrial** involvement, not reaching the pelvic wall
III	Lower third of vagina and/or pelvic wall and/or hydronephrosis/non-functioning kidney and/or pelvic/para-aortic nodes
IIIA	**Lower third of the vagina**, no pelvic wall extension
IIIB	**Pelvic side wall** and/or **hydronephrosis or non-functioning kidney** (unless known to be from another cause)
IIIC1	**Pelvic** lymph node metastasis (r = imaging, p = pathology)
IIIC2	**Para-aortic** lymph node metastasis
IV	Beyond the true pelvis or involving bladder/rectal mucosa
IVA	Spread to **mucosa of bladder or rectum** (biopsy-proven; **bullous oedema alone is not enough**)
IVB	**Distant** organs (lung, liver, bone, supraclavicular/inguinal nodes, peritoneum)

EXAM FAVOURITE

Changes in FIGO 2018: IB split into three by size (2 cm and 4 cm cut-offs); the **horizontal-spread criterion was removed** from stage IA; **lymph node involvement now upstages to IIIC** (imaging or pathology may be used, and pathology takes priority); micrometastases count as IIIC (2019 corrigendum), isolated tumour cells do not change the stage.

CLINICAL PEARL

When in doubt between two stages, **FIGO says assign the lower stage**. And remember the two traps: **hydronephrosis = IIIB regardless of tumour size**, and **bullous oedema of the bladder is not IVA**.

Performance status (ECOG): 0–2 can usually tolerate chemoradiation; 3–4 → often palliative intent.

6. Clinical Features

History

Ask	Why: the answer that changes the plan
Abnormal bleeding: post-coital, inter-menstrual, post-menopausal, contact bleeding; how long?	The classic early symptoms of invasive cancer; long duration + many symptoms usually means a bigger stage
Vaginal discharge: offensive, watery, blood-stained	Necrotic tumour: advanced disease; you will need metronidazole and a gentle examination
Pain: pelvic, low back, sciatic radiating down the leg	Parametrial/pelvic wall or nerve involvement → suspect **IIIB**
Leg swelling	Lymphatic/venous obstruction by pelvic wall disease; DVT

Ask	Why: the answer that changes the plan
Urine or faeces per vagina, haematuria, rectal bleeding, tenesmus	**Vesicovaginal or rectovaginal fistula** or bladder/rectal invasion (stage **IVA** or post-radiation) → **cystoscopy/proctoscopy**
Reduced urine output, vomiting, hiccups, itching, confusion	**Uraemia** from bilateral ureteric obstruction (IIIB), the commonest cause of death → urgent creatinine and renal ultrasound; cisplatin may be impossible
Weight loss, anorexia, cough, bone pain, jaundice, abdominal swelling	Distant spread (**IVB**) → CXR, abdominal ultrasound; palliative intent
Ability to self-care (ECOG)	ECOG 0–2 can usually tolerate chemoradiation; ECOG 3–4 → often palliative intent
Screening history (VIA/HPV/Pap, treatment received)	Never screened is the major risk factor; previous ablation/LEEP matters; a previous benign biopsy does not exclude cancer now
HIV status, ART regimen, adherence, last viral load and CD4	Risk, prognosis, tolerance of chemoradiation, drug interactions, need to optimise ART before treatment
Parity, age at first intercourse, number of partners, smoking	Risk factors
Pregnant? Desire for future fertility?	Pregnancy changes everything; fertility wish matters in stage IA–IB1 (cone, radical trachelectomy)
Comorbidity (diabetes, hypertension, heart disease, TB)	Fitness for radical surgery or cisplatin
Social: distance to Kampala, money, caregiver, what she knows and wants	Whether she can complete **5–7 weeks** of daily chemoradiation within **56 days** at UCI; early link to palliative care and social support

Examination

Follow the order of Hutchison's Clinical Methods (24th edition). In a heavily bleeding or uraemic woman, resuscitate first (section 9) and do the pelvic examination in theatre.

Introduction, consent and preparation

- Introduce yourself, confirm identity, explain, gain consent; **chaperone** for the pelvic examination; be gentle (friable tumour).
- **Hand hygiene**.
- **Position and exposure**: supine for the abdomen; lithotomy or dorsal with good light for the pelvis; keep her covered.

General inspection and vital signs

- **Pallor** (anaemia from bleeding), cachexia, jaundice, **performance status (ECOG)**, signs of **uraemia** (acidotic breathing, uraemic fetor, drowsiness).
- **Vitals**: pulse, BP, shock index if bleeding, temperature, RR.

Hands, face and neck

- Palmar and conjunctival pallor; **left supraclavicular (Virchow) lymph nodes** (IVB).

Chest

- Effusion, crackles (lung metastases).

Abdomen and groins

- Hepatomegaly, ascites, **ballotable/palpable kidneys** (hydronephrosis), pelvic mass (pyometra, bulky tumour), **inguinal nodes** (IVB).

- **Leg oedema** (unilateral: pelvic wall or DVT).

Pelvic examination

1. **Speculum**: describe the lesion: **exophytic** (cauliflower), **ulcerative** (crater with raised everted edges) or **endophytic** (hard barrel-shaped cervix, often adenocarcinoma); estimate **size in cm**; **vaginal extension**: none / upper two-thirds / **lower third**; contact bleeding, necrosis, offensive discharge.
2. **Biopsy** now if a lesion is visible: punch biopsy **from the edge** (viable tumour), not the necrotic centre; pack or apply Monsel's for haemostasis.
3. **Bimanual**: size and hardness of the cervix, mobility of the uterus, vaginal fornix involvement.
4. **Rectovaginal examination (index finger in the vagina, middle finger in the rectum)**: essential for staging. Feel each **parametrium** between the cervix and the pelvic side wall:
   - **Soft and free** → no parametrial involvement (stage I or IIA).
   - **Thickened, nodular, but a gap remains** between the thickening and the side wall → **IIB**.
   - **Thickened out to the side wall**, no free space ("frozen pelvis") → **IIIB**.
   - Check the **rectal mucosa** for tethering or ulceration (IVA needs biopsy).

Completing the examination

Thank her, cover her, hand hygiene, summarise.

Document: tumour size, vaginal extent, each parametrium (right and left separately), rectal mucosa, nodes, kidneys; e.g. "4 cm exophytic growth, upper vagina involved anteriorly, right parametrium thickened not reaching side wall, left free, rectal mucosa free".

EXAM FAVOURITE

Clinical FIGO staging exam, as done in Uganda: FIGO staging is still largely **clinical** in low-resource settings. **EUA** (examination under anaesthesia, ideally a gynae-oncologist and a radiation oncologist together) with **speculum, bimanual and rectovaginal examination**, **biopsy**, **cystoscopy** (and **proctoscopy/sigmoidoscopy**) if bladder or rectal involvement is suspected, plus assessment of the ureters by **renal ultrasound or IVU** (hydronephrosis or a non-functioning kidney = **IIIB**) and a **chest X-ray**. FIGO 2018 allows CT/MRI/PET and pathology to modify the stage where available, but you must always be able to stage with your hands.

Red flags

RED FLAG

- **Heavy vaginal bleeding** from the tumour → resuscitate, pack, tranexamic acid, urgent haemostatic radiotherapy.
- **Oliguria/anuria, rising creatinine** → obstructive uropathy (stage IIIB); needs ultrasound and discussion about nephrostomy/stent vs palliation.
- **Fever with foul discharge** → infected necrotic tumour or pyometra.
- **Leg swelling with pain** → DVT (cancer is prothrombotic) or pelvic wall disease.
- **Uncontrolled pain** → palliative emergency; start oral morphine.
- **Suicidal ideation/depression**, abandonment by partner, stigma: ask gently.

RED FLAG

Call a senior at once for: active heavy bleeding or shock index ≥0.9; anuria/oliguria or confusion (obstructive uropathy); fever with hypotension; or suspected spinal cord compression (new leg weakness, loss of bladder/bowel control).

Thinking like a clinician

- **Symptomatic women are not screened: they are diagnosed.** A woman with PCB and a VIA-negative cervix still needs a careful look and, if bleeding persists, colposcopy and biopsy. Adenocarcinoma can sit in the canal, invisible on VIA.
- **Biopsy the edge**, not the necrotic centre: take a **punch biopsy from the margin** of the growth, where viable tumour meets normal tissue. Necrotic tissue gives "necrosis: no viable tumour".
- **Do not ablate what might be cancer.** Ablation destroys the tissue that would have given the diagnosis and under-treats invasive disease: suspicious lesions go for biopsy.
- **Stage the patient, not the scan**: FIGO 2018 allows imaging and pathology, but most Ugandan women are staged by examination ± ultrasound of the kidneys (hydronephrosis alone = IIIB) and a chest X-ray.
- **Match treatment to stage and to the patient**: a 35-year-old with a 1.5 cm tumour and good performance status can be cured by surgery; a 60-year-old with IIIB disease and creatinine 600 µmol/L may be best served by symptom control. Ask the patient what she understands and wants.
- **Always think about the kidneys.** Check urea, creatinine and renal ultrasound before chemoradiation: cisplatin is nephrotoxic and cannot be given if creatinine clearance is poor.
- **Cancer in pregnancy** (not rare in Uganda): management depends on stage and gestation; discuss at a multidisciplinary level; delivery is by **caesarean section** (vaginal delivery through a tumour risks haemorrhage and tumour implantation in the episiotomy).
- **Common student mistakes**: calling an ulcerated cervix "cervicitis" and giving antibiotics; forgetting the rectal examination; forgetting renal function; staging hydronephrosis as stage II; offering cryotherapy to a lesion that looks invasive; forgetting palliative care and pain relief.

7. Differential Diagnoses

Differential	Distinguishing clinical feature	Distinguishing investigation
Cervical cancer	Hard, friable, exophytic or ulcerated growth; contact bleeding; parametrial thickening	**Punch biopsy**
Cervical ectropion	Flat, regular, symmetrical red area around os; young/pregnant/COC	VIA/HPV negative; reassurance
Cervicitis	Mucopus, oedema, friability; resolves with antibiotics	NAAT/culture; re-examine after treatment
Cervical polyp	Smooth pedunculated mass, stalk visible	Avulsion + histology
Prolapsed submucous fibroid	Firm, pale, smooth mass with stalk through os	Ultrasound; EUA
Endometrial cancer extending to cervix	Post-menopausal, obese, diabetic; bulky uterus	Endometrial biopsy; histology (adenocarcinoma type)
Tuberculous / schistosomal cervicitis	Ulcerative or granular lesions; endemic areas; infertility	Biopsy (granulomas, ova), AFB/GeneXpert
Chancroid / syphilitic chancre / herpes on cervix	Painful (chancroid, HSV) or painless (syphilis) ulcer; partner history	RPR/TPHA, swab; biopsy if not healing
Condylomata acuminata	Warty lesions (HPV 6/11), often multiple, vulval too	Biopsy if atypical, especially in HIV
Products of conception at the os	Positive hCG, recent pregnancy, open os	hCG, ultrasound

8. Clinical Workup

The workup of a woman with suspected cervical cancer has three jobs, done in this order: **(1) keep her alive today** (bleeding, uraemia, sepsis; section 9), **(2) prove the diagnosis** (biopsy) and **(3) stage her and check her fitness** so that the multidisciplinary team at UCI/Mulago can choose surgery, chemoradiation or palliation. At HC IV or a general hospital you can do almost all of this before referral, and doing it well saves her weeks.

Bedside

"Available at" is the lowest level that usually offers the test Verify *availability compiled from the source notes, not an MoH list*.

Test	What it tells you	Available at
Urine pregnancy test	Pregnancy changes the whole plan (MDT, timing, caesarean delivery)	All levels
HemoCue / Hb	Anaemia is almost universal; **Hb <7 g/dL** → transfuse; target **Hb ≥10 g/dL** during radiotherapy	HC III and above
Urine dipstick	Haematuria (bladder invasion or radiation cystitis), UTI	All levels
Urine output (catheter if unwell)	Oliguria → obstructive uropathy	All levels
Random blood sugar	Diabetes (steroids with chemotherapy, surgical risk)	HC III and above
Rapid HIV test	If status unknown; HIV changes treatment tolerance and ART planning	All levels
Point-of-care ultrasound of the kidneys	**Hydronephrosis in minutes**: if present (and no other cause), the stage is **at least IIIB**	HC IV and above with an operator
VIA (only if no obvious growth)	A frank growth needs a **biopsy**, not VIA; VIA is for women without a visible lesion	HC III (trained) and above

Screening and diagnosis of pre-cancer

Test	How / what you look for	Notes
VIA	Apply **3–5% acetic acid**, wait **1 minute**; **positive** = dense, well-defined **acetowhite** lesion touching/adjacent to the SCJ; **suspicious for cancer** = growth/ulcer	Cheap, immediate, allows same-visit treatment; subjective; poor after menopause (SCJ inside canal)
HPV DNA test (clinician- or **self-collected** vaginal sample; GeneXpert/careHPV)	High-risk HPV positive/negative (± genotype 16/18)	**WHO preferred primary test**; very high negative predictive value → long intervals
Cytology (Pap/LBC)	Dyskaryosis (ASC-US, LSIL, HSIL)	Needs lab and pathologist; used mostly in private/tertiary settings
Colposcopy	Magnified view: acetowhite, mosaicism, punctation, atypical vessels; guides biopsy	Referral hospitals
Punch biopsy	Histology: CIN grade or invasion	Required for any suspected cancer
LEEP/cone specimen	Histology of the whole TZ, margins	Diagnostic and therapeutic

Basic lab

Test	When	Expected / abnormal finding	How it changes management	Available at
FBC	All	Anaemia; high WBC (infection); low platelets	Transfuse to **Hb ≥10 g/dL** for radiotherapy (hypoxic tumour is radioresistant); chemotherapy needs adequate counts	HC IV and above
Urea, creatinine, electrolytes (± eGFR/creatinine clearance)	All, and **weekly during cisplatin**	Raised creatinine = obstructive uropathy (IIIB) or other renal disease; low Mg/K on cisplatin	**Cisplatin is withheld if creatinine clearance is below about 50 mL/min** (carboplatin may be substituted); **drain the kidneys first** (nephrostomy) if she is for curative treatment	HC IV (some), general hospitals and above
HIV test, CD4, viral load	All (known positive: CD4 and VL)	Unsuppressed VL, low CD4	Optimise ART before and during chemoradiation; co-trimoxazole prophylaxis per HIV guidelines	HIV clinics; CD4/VL via hub samples
LFTs, RBS, hepatitis B	Before chemotherapy/surgery	Raised enzymes, diabetes	Fitness, drug choice	General hospital and above
Group and cross-match	Anaemic, bleeding, or before surgery	Compatible units	Blood ready for transfusion/surgery	HC IV and above

Imaging

Test	When	Expected / abnormal finding	How it changes management	Available at
Renal ultrasound	All	**Hydronephrosis** (uni- or bilateral), non-functioning kidney	**Upstages to IIIB** (unless another cause); nephrostomy discussion	Most general hospitals and many HC IVs
Chest X-ray	All	Lung metastases, effusion	**IVB** → palliative intent	Widely available
Abdominal ultrasound	All	Liver metastases, ascites, para-aortic nodes	IVB / IIIC2 suspicion	General hospital and above
IVU (intravenous urogram)	Where ultrasound is inconclusive and renal function allows	Ureteric obstruction, non-functioning kidney	Same staging meaning as hydronephrosis (IIIB); contrast is risky if creatinine is high	Regional/national referral
MRI pelvis	Where available, especially if surgery is planned	Tumour size, parametrial invasion, nodes	Best for size and parametrium; separates IB2 (surgery) from IIB (chemoradiation)	UCI, Mulago, private centres
CT chest/abdomen/pelvis	Locally advanced disease, before radiotherapy planning	Pelvic nodes (**IIIC1r**), para-aortic nodes (**IIIC2r**), distant metastases	Extended-field radiotherapy for para-aortic nodes; IVB → palliation	UCI, Mulago, regional centres, private

Specialised / referral-level

Test	When	Expected / abnormal finding	How it changes management	Available at
Punch biopsy (histology)	Every visible or suspicious lesion, **before any treatment**	Squamous cell carcinoma (about 75–80%), adenocarcinoma (about 15–20%), other	No treatment without histology	Biopsy at HC IV/general hospital; histology at regional/national referral labs (Mulago, UCI, Mbarara, Gulu, Lacor and others); turnaround often 1–3 weeks, so send early
EUA with cystoscopy ± proctoscopy	Locally advanced disease, urinary or rectal symptoms, or clinical staging needed	Bladder/rectal **mucosal** tumour (biopsy-proven)	**IVA** (bullous oedema alone does **not** make IVA)	Theatre at general/regional/national referral hospitals
Colposcopy ± LEEP	Persistent PCB with negative VIA; suspected glandular lesion; not eligible for ablation	CIN grade, invasion, margins	Diagnosis and treatment of pre-cancer	Referral hospitals

UGANDA CONTEXT

At a **general hospital or HC IV** you can do the core staging set before referral: **biopsy, Hb, creatinine, renal ultrasound, CXR and HIV status**. Sending her to UCI with these results (and the histology report) avoids repeat tests and speeds treatment. MRI/CT, cystoscopy under EUA and radiotherapy planning are done at **UCI/Mulago** and a few regional cancer centres.

Putting it together

Model one-line summary: "A 48-year-old P6, HIV-positive on TLD with suppressed viral load, never screened, with 5 months of post-coital bleeding and offensive discharge; a 4.5 cm exophytic cervical growth involving the upper vagina, right parametrium thickened not reaching the side wall, left free; Hb 8.9 g/dL, creatinine 82 µmol/L, kidneys normal on ultrasound, CXR clear; biopsy: squamous cell carcinoma."

Problem list

1. Invasive squamous cell carcinoma of the cervix, **FIGO IIB** (clinical).
2. Anaemia (Hb 8.9 g/dL): below the radiotherapy target.
3. HIV on ART (suppressed): continue ART; check CD4.
4. Offensive discharge and pain: symptom control.
5. Social: distance to UCI, money, caregiver.

Working diagnosis and plan: cervical cancer FIGO stage IIB → refer to UCI for **concurrent chemoradiation** (EBRT 45–50 Gy in 25 fractions + weekly cisplatin 40 mg/m² + brachytherapy, total ≤56 days); transfuse to Hb ≥10; metronidazole for odour; analgesia; link to palliative care and counselling.

Model case presentation (ward round)

> "Mrs K is a 52-year-old P7, HIV-positive on TLD with suppressed viral load, never screened, who presents with 6 months of post-coital and then continuous offensive blood-stained vaginal discharge, left sciatic pain and weight loss. She is pale (Hb 7.4 g/dL). Speculum shows a 5 cm fungating, friable exophytic growth on the cervix extending onto the upper vagina; rectovaginal examination shows the left parametrium thickened and fixed to the pelvic side wall. Renal ultrasound shows left hydronephrosis; creatinine is 110

> μmol/L; CXR is clear. Punch biopsy: moderately differentiated squamous cell carcinoma. My working diagnosis is cervical cancer FIGO stage IIIB (pelvic wall and hydronephrosis). Plan: transfuse to Hb ≥10, refer urgently to UCI for concurrent chemoradiation (EBRT 45–50 Gy with weekly cisplatin 40 mg/m², renal function permitting, followed by brachytherapy), start oral morphine for pain with a laxative, metronidazole for odour, link to palliative care, and counsel her and her family."

Worked example 1: staging from the examination

> A 42-year-old: 3 cm exophytic tumour confined to the cervix; vagina free; both parametria soft on rectovaginal examination; kidneys normal on ultrasound; CXR clear.

Finding	Stage reasoning
Visible tumour, 3 cm, confined to the cervix	Stage IB (visible = at least IB); **≥2 and <4 cm = IB2**
Parametria free, vagina free	Not II
No hydronephrosis, CXR clear	Not IIIB/IVB

- **Stage IB2** → options: **radical hysterectomy + pelvic lymphadenectomy** (open approach) or **primary chemoradiation**. If MRI/CT or surgery later shows **positive pelvic nodes** she becomes **IIIC1r** (imaging) or **IIIC1p** (pathology) and needs chemoradiation.

Worked example 2: how one finding changes the stage

Same 4 cm tumour, plus…	FIGO 2018 stage	Treatment intent
Nothing else, confined to the cervix	**IB3** (≥4 cm)	Chemoradiation (preferred)
Upper-vaginal involvement, parametria free	**IIA2**	Chemoradiation
Right parametrium thickened, **gap** to side wall	**IIB**	Chemoradiation
Tumour reaches the **lower third of the vagina**	**IIIA**	Chemoradiation
Parametrium **fixed to the side wall**, or **hydronephrosis** on ultrasound (even with free parametria)	**IIIB**	Chemoradiation if kidneys can be drained and renal function allows; otherwise palliation
Enlarged **pelvic nodes** on CT	**IIIC1r**	Chemoradiation
Enlarged **para-aortic nodes** on CT	**IIIC2r**	Extended-field chemoradiation
Biopsy-proven **bladder mucosa** tumour at cystoscopy	**IVA**	Chemoradiation (fistula risk) or palliation
Lung or liver metastases, supraclavicular node	**IVB**	Palliative chemotherapy/RT and supportive care

Worked example 3: can she have cisplatin?

> A 55-year-old, weight 50 kg, creatinine 150 μmol/L, bilateral hydronephrosis.

- **Cockcroft–Gault** (women): creatinine clearance ≈ (140 − age) × weight (kg) × 1.04 ÷ creatinine (μmol/L) = 85 × 50 × 1.04 ÷ 150 ≈ **29 mL/min**.
- Below about 50 mL/min → **no cisplatin** as it stands. Stage is **IIIB** (hydronephrosis). Options: **nephrostomy** then re-check renal function and proceed with chemoradiation, or radiotherapy alone/carboplatin per the oncologist, or palliation if frail and that is her wish. Exact cut-offs vary between oncology units.

9. Management

Immediate / emergency

First 5 minutes (ABCDE)

1. **A, B, C and vitals**: pulse, BP, **shock index** (HR ÷ SBP; **≥0.9** abnormal, **≥1.4** needs urgent intervention), RR, temperature, level of consciousness.
2. **Is she bleeding heavily?** (pads soaked, clots, pallor, faintness). If yes: **call for help**, **two large-bore IV cannulae**, blood for **Hb and group and cross-match**, crystalloid, **tranexamic acid 1 g IV**, and a **tight vaginal pack** with a urinary catheter (algorithm below). Transfuse as needed.
3. **D: is she uraemic?** Drowsy or confused, acidotic (deep sighing) breathing, vomiting, hiccups, itching, little urine → catheterise and measure output, send **urea, creatinine and potassium urgently**, and do a **renal ultrasound** today. Hyperkalaemia and fluid overload can kill before the cancer does.
4. **E: is she septic?** Fever ≥38.0 °C with foul discharge (infected necrotic tumour, pyometra) or a urinary source → cultures if possible and IV antibiotics (e.g. ceftriaxone + metronidazole) per local protocol.
5. **Is she in severe pain?** Start analgesia now (paracetamol, then oral morphine 2.5–5 mg every 4 hours, lower/less often in renal failure); do not make her wait for the diagnosis.
6. **Pregnancy test** in every woman of reproductive age.

Heavy bleeding from cervical cancer (algorithm)

1. **Call for help**, ABC, **two large-bore IV cannulae**, bloods for Hb and **group and cross-match**, start crystalloid; **transfuse** as needed.
2. **Tranexamic acid 1 g IV** 8-hourly (then 1 g PO 8-hourly).
3. **Tight vaginal packing** with gauze (soaked in saline, or in Monsel's solution/adrenaline where available); insert a **urinary catheter** (packing obstructs the urethra). Leave pack 24–48 h; give antibiotics (e.g. metronidazole) to reduce infection and odour.
4. **Urgent haemostatic radiotherapy** (short hypofractionated course, e.g. a single large fraction): refer to UCI.
5. If bleeding continues and facilities exist: **uterine/internal iliac artery embolisation or ligation**.
6. Once stable: correct anaemia, stage, plan definitive or palliative treatment.

Obstructive uropathy (renal failure)

- Bilateral hydronephrosis with rising creatinine = **stage IIIB**, the **commonest cause of death** in cervical cancer.
- If she is a candidate for curative chemoradiation or wants active treatment: **percutaneous nephrostomy** (interventional radiology) or ureteric stenting (often impossible because of tumour), then treatment; dialysis is rarely feasible.
- If disease is advanced and she is frail, many women choose **palliation**: uraemia may be a peaceful death if pain, nausea and itching are controlled. This must be a **shared, honest decision with the patient and family**.

Definitive

> **NOTE**
> Treatment is decided by a **multidisciplinary team** (gynae-oncologist, radiation and medical oncologist, pathologist, radiologist, palliative care), in Uganda usually at **UCI/Mulago**. Your role at HC IV or a general hospital: **suspect, biopsy, basic staging (renal US, creatinine, CXR, Hb), refer promptly, and start palliation of symptoms**.

A. Pre-cancer: the screen-and-treat visit, step by step (HC IV / HIV clinic)

1. **Counsel and consent**: explain HPV, the test, what a positive result means and the treatment options; offer HIV testing.
2. **Take the sample** (HPV self/clinician sample) or perform **VIA**: speculum, identify the SCJ, apply 3–5% acetic acid, wait 1 minute, inspect.
3. **Classify**:
   - **VIA/HPV negative** → re-screen at the recommended interval (section 12).
   - **HPV positive** → **triage with VIA** (especially in WLWH) and assess eligibility for ablation.
   - **VIA positive**, lesion **eligible for ablation** → treat the same day.
   - **Not eligible for ablation** → **LEEP** (referral hospital).
   - **Suspicious for cancer** → **biopsy** and refer: do **not** ablate.
4. **Eligibility for ablation**: the whole lesion and the **SCJ fully visible** (TZ type 1 or 2), lesion covers **<75%** of the ectocervix, does **not extend into the canal** or onto the vagina, and there is **no suspicion of invasive or glandular disease**. Not during pregnancy (defer to 6 weeks postpartum; treat only if cancer is suspected, then biopsy, not ablation).
5. **Treat**:
   - **Thermal ablation**: reusable probe heated to **at least 100 °C** applied for **20–30 seconds** per application, overlapping to cover the whole TZ; takes minutes, needs only electricity/battery, no gas; WHO-preferred for scale-up.
   - **Cryotherapy**: CO_2 or N_2O gas; **double-freeze technique: 3 minutes freeze, 5 minutes thaw, 3 minutes freeze**, ensuring an ice-ball extends 4–5 mm beyond the lesion.
   - **LEEP/LLETZ**: under local anaesthesia (lignocaine with adrenaline), excise the TZ with a wire loop; gives a histology specimen; haemostasis with ball diathermy/Monsel's.
6. **After-care advice**: watery discharge for 2–4 weeks; **no intercourse or tampons for 4 weeks** (or condoms); return for heavy bleeding, fever, offensive discharge or severe pain.
7. **Follow-up**: re-screen at **12 months**; HIV-positive women continue shorter intervals.

> CLINICAL PEARL
> **Why ablation is enough for CIN**: CIN is confined to the epithelium and the crypts (up to about 5 mm deep). Ablation destroys to that depth, so the diseased TZ is replaced by healthy squamous epithelium. It cannot treat invasion, which is why suspected cancer must be excluded first.

B. CIN found on histology (referral level)

Finding	Management
CIN 1	Usually regresses: **follow-up** (repeat test at 12 months); treat if persistent ≥2 years or in WLWH per local policy
CIN 2–3	**Treat**: LEEP (preferred when available, gives margins) or ablation if eligible
AIS	**Excision** (cone/LEEP with clear margins), not ablation; hysterectomy when family complete
Pregnancy with CIN	Colposcopy to exclude invasion; **defer treatment** until ≥6 weeks postpartum

C. Invasive cancer by stage

Stage	Standard treatment	Notes / fertility-sparing
IA1 (no LVSI)	**Cone biopsy/LEEP with clear margins** (fertility) or **simple (extrafascial) hysterectomy**	Node risk <1%

Stage	Standard treatment	Notes / fertility-sparing
IA1 with LVSI, IA2	**Modified radical hysterectomy + pelvic lymphadenectomy** (or cone with clear margins + pelvic nodes if fertility desired)	Radical trachelectomy in selected young women
IB1, IB2, IIA1	**Radical (Wertheim-type) hysterectomy + bilateral pelvic lymphadenectomy or** primary **chemoradiation** (equal cure rates; surgery preserves ovarian function and vaginal elasticity in young women)	Radical trachelectomy for IB1 (≤2 cm) wanting fertility; open approach preferred (LACC trial showed worse survival with minimally invasive radical hysterectomy)
IB3, IIA2	**Concurrent chemoradiation** (preferred)	Avoid surgery + radiation "double treatment"
IIB, III (A, B, C1), IVA	**Concurrent chemoradiation**: EBRT + weekly cisplatin + brachytherapy	Extended-field RT if para-aortic nodes (IIIC2)
IVB	**Palliative systemic chemotherapy** (carboplatin/cisplatin + paclitaxel) and/or **palliative RT** to symptomatic sites; **best supportive care**	Performance status and wishes guide intensity

Radical hysterectomy (Wertheim/Meigs; Querleu-Morrow type C) removes the uterus, cervix, **parametria**, **upper third (about 2 cm) of the vagina** and pelvic lymph nodes; the **ovaries can be conserved** in young women with squamous cancer. The ureters are dissected out of their tunnels: risks include **ureteric injury/fistula**, **bladder dysfunction** (autonomic nerve damage), lymphocyst and lymphoedema.

Adjuvant treatment after radical hysterectomy

- **High-risk** (Peters criteria): **positive nodes, positive parametria or positive margins → chemoradiation**.
- **Intermediate-risk** (Sedlis criteria: combinations of LVSI, deep stromal invasion, tumour size) → **pelvic radiotherapy**.

Chemoradiation: what it involves (UCI practice guideline)

Component	Regimen	Why
External beam RT (EBRT)	**45–50 Gy in 25 fractions** (daily, Monday–Friday, 5 weeks) to the pelvis	Treats tumour, parametria and pelvic nodes
Concurrent cisplatin	**40 mg/m² IV weekly × 5–6 cycles** (with pre-hydration and antiemetics); check creatinine each week	**Radiosensitiser**: improves survival by about 10% over RT alone
Brachytherapy (HDR)	e.g. **24–28 Gy in 3–4 fractions** to the tumour (intracavitary tandem and ovoids/ring ± interstitial needles)	Delivers a very high dose to the cervix with rapid fall-off, sparing bladder and rectum
Overall treatment time	**≤56 days (8 weeks)**	Every day of delay beyond this lets tumour cells repopulate and reduces local control
Supportive	Keep **Hb ≥10 g/dL**, nutrition, ART, antiemetics, manage diarrhoea/cystitis	Anaemia → hypoxic tumour → radioresistance

D. Palliative care: the reality for most Ugandan patients

Symptom	Management
Pain	WHO analgesic ladder: **paracetamol 1 g PO 6–8-hourly** ± NSAID (e.g. ibuprofen 400 mg 8-hourly if renal function allows) → **oral morphine** (immediate-release solution) starting **2.5–5 mg every 4 hours** with the same dose for breakthrough, titrated daily; **always prescribe a laxative** (e.g. bisacodyl) with morphine; reduce dose/frequency in renal failure. Neuropathic pain (sciatic): add **amitriptyline 12.5–25 mg nocte**. Bone pain: palliative RT (single 8 Gy fraction)
Bleeding	Tranexamic acid 1 g PO 8-hourly; vaginal packing; **palliative/haemostatic RT**
Offensive discharge	**Metronidazole 400 mg PO 8-hourly** (or crushed metronidazole tablets applied topically/in vaginal packs), vaginal douching with saline, dressings, charcoal
Fistula (VVF/RVF)	Pads, barrier creams (zinc oxide), catheter; urinary diversion or colostomy in selected patients
Leg lymphoedema	Elevation, compression, skin care; treat cellulitis
Nausea (uraemia/opioids)	Haloperidol 1.5–3 mg nocte or metoclopramide 10 mg 8-hourly
Psychosocial and spiritual	Explain diagnosis honestly, involve family, address stigma and abandonment, link to **hospice** (e.g. Hospice Africa Uganda, Mulago palliative care unit) and home-based care

UGANDA CONTEXT

In Uganda, **oral liquid morphine is available free** in public facilities, and specially trained **nurses and clinical officers can prescribe it**. Fear of addiction is common among patients and staff: reassure that morphine for cancer pain, correctly titrated, is safe. Every woman diagnosed with advanced cervical cancer should be **linked to palliative care on the day of diagnosis**, in parallel with any oncological treatment.

Drug table

Drug	Dose (adult)	Route	Frequency	Duration	Key cautions
Cisplatin (with RT)	**40 mg/m²**	IV, with pre-hydration (normal saline) and antiemetics	Weekly	5–6 cycles during EBRT	Nephrotoxic, ototoxic, highly emetogenic, neuropathy, myelosuppression; check creatinine, Mg, K weekly; **withhold if creatinine clearance below about 50 mL/min** (carboplatin may be substituted); not until obstructed kidneys are drained

Drug	Dose (adult)	Route	Frequency	Duration	Key cautions
Carboplatin / cisplatin + paclitaxel	Per oncologist	IV	Cycles	Per oncologist	Palliative chemotherapy for IVB
Ondansetron + dexamethasone	Per oncology protocol	IV/PO	With cisplatin	Each cycle	Antiemetic cover
Tranexamic acid	**1 g**	IV, then PO	8-hourly	Until bleeding controlled	Tumour bleeding; thrombosis risk
Metronidazole	**400 mg** (or crushed tablets topically / in packs)	PO / topical	8-hourly	Ongoing for odour	Avoid alcohol
Ceftriaxone + metronidazole	Per local protocol	IV	Per protocol	Until afebrile, then oral	Sepsis from necrotic tumour, pyometra, urinary source
Paracetamol	**1 g**	PO	6–8-hourly	Ongoing	Max 4 g/day
Ibuprofen	400 mg	PO	8-hourly	As needed	Only if renal function allows
Oral morphine (immediate-release solution)	**2.5–5 mg** start (opioid-naive), same dose for breakthrough; titrate daily	PO	Every 4 hours	Ongoing	**Always prescribe a laxative**; reduce dose/frequency in renal failure
Bisacodyl	Per BNF/UCG	PO	Nocte	While on morphine	Opioid constipation
Amitriptyline	**12.5–25 mg**	PO	Nocte	Ongoing	Neuropathic (sciatic) pain; sedation
Haloperidol	**1.5–3 mg**	PO	Nocte	As needed	Nausea from uraemia/opioids
Metoclopramide	**10 mg**	PO/IV	8-hourly	As needed	Nausea
Lignocaine with adrenaline	Local infiltration	Cervical	Once	At LEEP	Local anaesthesia for LEEP

Drug	Dose (adult)	Route	Frequency	Duration	Key cautions
HPV vaccine	**0.5 mL**	IM deltoid	Single dose (Uganda, 10-year-old girls); ≥2 (ideally 3) doses if immunocompromised	n/a	Not a substitute for screening

DRUG DOSING

Cisplatin is nephrotoxic, ototoxic and highly emetogenic: pre-hydrate with normal saline, give antiemetics (e.g. ondansetron + dexamethasone), check creatinine/electrolytes (magnesium, potassium) weekly, and withhold if creatinine clearance is below about 50 mL/min; carboplatin may be substituted. Do not start cisplatin in a patient with obstructive uropathy until the kidneys are drained.

Referral criteria

Referral triggers compiled from the source notes and UCG levels of care Verify *not an MoH referral list*.

Level	What to do	Refer up when
HC III	Counsel, VIA (where trained), HPV sample collection, symptom history	VIA positive, HPV positive, any visible lesion, PCB/PMB/offensive discharge → HC IV
HC IV / HIV clinic	Screen-and-treat (thermal ablation/cryotherapy), punch biopsy, Hb, rapid HIV, renal POCUS, oral morphine, start palliation	Not eligible for ablation (LEEP), biopsy needed for histology, suspected invasive cancer, heavy bleeding not controlled by packing → general/regional referral hospital
General / regional referral hospital	Core staging set (biopsy, Hb, creatinine, renal ultrasound, CXR, HIV status), EUA ± cystoscopy, LEEP, colposcopy, transfusion, packing	Confirmed invasive cancer for MDT, surgery or chemoradiation; haemostatic RT; nephrostomy → UCI/Mulago (or regional cancer centres)
UCI / Mulago	MDT, MRI/CT, radical hysterectomy, chemoradiation, brachytherapy, palliative care unit, hospice link	n/a

Monitoring and follow-up

- **During chemoradiation**: creatinine and electrolytes (Mg, K) weekly, FBC, hearing, emesis; keep **Hb ≥10 g/dL**; ART adherence; complete within **56 days**.
- **After treatment of pre-cancer**: re-test at **12 months**; WLWH continue shorter intervals.
- **After cancer treatment**: every **3–6 months for 2 years**, then 6–12-monthly to 5 years: symptoms, speculum and rectovaginal examination; imaging only if symptoms.
- **Vaginal dilators** after radiotherapy (prevent stenosis, allow examination and intercourse).
- Manage **menopausal symptoms** after RT (ovarian failure): HRT is not contraindicated in squamous cervical cancer.
- Watch for late RT effects: radiation proctitis/cystitis, fistula, vaginal stenosis, bowel strictures.
- **Palliative patients**: regular review of pain (titrate morphine daily), bleeding, odour, nausea, renal function only if it changes management; home-based care.

10. Complications

Disease-related

Complication	Cause / notes
Renal failure (obstructive uropathy)	Parametrial/pelvic wall spread compressing the ureters: **commonest cause of death**
Haemorrhage	Tumour erosion of vessels; can be torrential
Fistulae	**Vesicovaginal** (urine leak), **rectovaginal** (faeces); from tumour invasion
Anaemia	Chronic bleeding, malnutrition
Pyometra / sepsis	Tumour occluding the canal; infected necrotic tumour
Pain	Nerve, bone and pelvic wall involvement
Lymphoedema, DVT/PE	Pelvic node and vein involvement; hypercoagulability
Cachexia, malnutrition	Advanced disease
Psychosocial	Stigma (offensive odour, fistula), marital abandonment, financial ruin, depression

Treatment-related

Treatment	Complications
Radical hysterectomy	Ureteric/bladder injury, ureteric fistula, bladder dysfunction (autonomic nerve damage), lymphocyst, lymphoedema
Radiotherapy	Cystitis, proctitis, diarrhoea, vaginal stenosis, fistula (radiation necrosis), ovarian failure, bowel strictures
Cisplatin	Nephrotoxicity, ototoxicity, neuropathy, emesis, myelosuppression, hypomagnesaemia/hypokalaemia; anaemia
Ablation / LEEP	Watery discharge, bleeding, infection; LEEP/cone: stenosis, cervical insufficiency
Morphine	Constipation (always a laxative), nausea, drowsiness; accumulation in renal failure

11. Prognosis

Approximate 5-year survival with treatment: stage I **80–95%**, II **60–75%**, III **30–45%**, IVA **15–20%**, IVB **<10%**. In Uganda, overall survival is much lower because of late presentation (about 80% stage IIB or worse at UCI) and incomplete treatment (only 12% completing the full course in one UCI study). Every day of delay beyond 56 days of chemoradiation reduces local control. The commonest cause of death is renal failure from ureteric obstruction.

12. Prevention

WHO **90-70-90 by 2030**: **90%** of girls fully vaccinated with HPV vaccine by 15; **70%** of women screened with a high-performance test by 35 and again by 45; **90%** of women with pre-cancer treated and 90% with invasive cancer managed. Elimination threshold: **<4 per 100,000** women.

Primary prevention: HPV vaccination

Item	Details
Target	Girls aged **9–14** (Uganda: **10-year-old girls**, through schools and health facilities); WHO secondary targets include older girls/young women (catch-up), and boys where resources allow
Vaccine	Uganda uses the quadrivalent vaccine (HPV 6, 11, 16, 18); bivalent and nonavalent vaccines also exist

Item	Details
Schedule	WHO 2022: **1 or 2 doses** for girls 9–14 and young women 15–20; Uganda moved to a **single dose in 2025** (previously 2 doses 6 months apart)
Immunocompromised / HIV	**At least 2 doses, ideally 3**
Route	0.5 mL **IM** into the deltoid
Key message	Vaccination does **not** replace screening later in life

Secondary prevention: screening and treatment of pre-cancer (WHO 2021)

Population	Primary test	Start age	Interval	Approach	After treatment
General population	**HPV DNA** (preferred); VIA or cytology where HPV is not yet available	**30**	**5–10 years** (HPV); **3 years** (VIA/cytology)	Screen-and-treat (or screen–triage–treat)	Re-test at **12 months**
Women living with HIV	**HPV DNA**	**25**	**3–5 years** (HPV); 3 years (VIA/cytology)	**Screen, triage and treat**	Re-test at **12 months**

Screening can stop after 50 (general) or 65 (WLWH) if the last two screens were negative. **Self-sampling** for HPV is recommended as an option: it improves uptake in communities where women avoid speculum examinations. The screen-and-treat procedure is in section 9.

> UGANDA CONTEXT
> Uganda's cervical cancer programme began with **VIA-based "see-and-treat"** (with cryotherapy) for women aged **25–49**, with screening **every 3 years for HIV-negative** women and **annually for women living with HIV** under earlier national plans. Following the **WHO 2021 guideline**, Uganda is scaling up **HPV DNA testing** (including **self-sampling** and **GeneXpert** platforms at hubs) with **VIA triage** and **thermal ablation**, and the HIV consolidated guidelines now name HPV testing as the preferred primary test for women living with HIV. Check the current MoH algorithm at your facility: intervals are in transition.

Tertiary prevention: early diagnosis of symptomatic women (look and biopsy), prompt referral with the core staging set, completing chemoradiation within 56 days, early palliative care.

13. Special Populations

- **Women living with HIV**: about 6 times the risk; screen from 25, every 3–5 years, screen–triage–treat, re-test at 12 months after treatment; larger lesions, more recurrence, faster progression; optimise ART (suppressed viral load) and co-trimoxazole before and during chemoradiation; at least 2 (ideally 3) doses of HPV vaccine if a girl is HIV-positive.
- **Pregnancy**: CIN: colposcopy to exclude invasion and **defer treatment** until ≥6 weeks postpartum; do not ablate in pregnancy. Invasive cancer: management depends on stage and gestation, decided by the MDT; deliver by **caesarean section** (vaginal delivery through a tumour risks haemorrhage and tumour implantation in the episiotomy).
- **Young women wanting fertility**: IA1 cone/LEEP with clear margins; IA2–IB1 radical trachelectomy in selected women; ovaries can be conserved at radical hysterectomy in squamous cancer.
- **Post-menopausal women**: SCJ inside the canal, so VIA is poor; HPV testing or colposcopy preferred.
- **Renal impairment**: no cisplatin below a creatinine clearance of about 50 mL/min; drain the kidneys first; reduce morphine dose and frequency.

14. Key Points

High-yield points

- Persistent high-risk HPV is necessary: **E6 degrades p53, E7 inactivates Rb**.
- **Symptomatic women are diagnosed, not screened**: look and biopsy the **edge** of any growth.
- **Never ablate a lesion that looks invasive.**
- Stage with your hands: **rectovaginal examination** separates IIB from IIIB.
- **Hydronephrosis = IIIB** whatever the size; **bullous oedema is not IVA**.
- Early disease: radical hysterectomy or chemoradiation; locally advanced (IB3–IVA): **chemoradiation within 56 days**.
- **Check the kidneys before cisplatin**; drain obstructed kidneys first.
- Keep **Hb ≥10 g/dL** during radiotherapy.
- Renal failure from ureteric obstruction is the **commonest cause of death**.
- Link every advanced case to **palliative care on the day of diagnosis**; oral morphine is free and safe.

Numbers & doses

Item	Value
HPV 16 and 18	Cause about **70%** of cervical cancers
HPV clearance	About **90%** of infections cleared within **1–2 years**
CIN → invasion	Usually **10–20 years** (faster with HIV)
HIV	About **6 times** the risk of cervical cancer
Uganda incidence / mortality	About **54 / 41 per 100,000** women (GLOBOCAN 2022)
Screening (WHO 2021), general population	From **30**, HPV DNA every **5–10 years** (VIA/cytology every 3 years); can stop after **50** if the last two screens were negative
Screening (WHO 2021), women living with HIV	From **25**, HPV DNA every **3–5 years**; screen, triage and treat; can stop after **65** if the last two were negative
After treatment of pre-cancer	Re-test at **12 months**
VIA	**3–5% acetic acid**, read at **1 minute**
Thermal ablation	**≥100 °C**, **20–30 s** per application
Cryotherapy	**3 min freeze – 5 min thaw – 3 min freeze**; ice-ball **4–5 mm** beyond the lesion
Ablation cut-off	Lesion **<75%** of the ectocervix, SCJ fully visible, no canal extension, no suspicion of cancer
After ablation	Discharge **2–4 weeks**; no intercourse or tampons **4 weeks**
HPV vaccine (Uganda)	**Single dose** for **10-year-old** girls (since 2025); HIV/immunocompromised **at least 2 doses** (ideally 3); 0.5 mL IM deltoid
WHO targets	**90-70-90** by 2030; elimination threshold **<4 per 100,000** women
FIGO IA1 / IA2	Invasion **<3 mm** / **≥3 and <5 mm**
FIGO IB1 / IB2 / IB3	**<2 cm** / **≥2 and <4 cm** / **≥4 cm**
FIGO IIA1 / IIA2	**<4 cm** / **≥4 cm** (upper two-thirds of vagina)
FIGO IIIB	Pelvic side wall **or hydronephrosis/non-functioning kidney**
FIGO IIIC1 / IIIC2	Pelvic nodes / para-aortic nodes

Item	Value
EBRT	**45–50 Gy in 25 fractions** (5 weeks)
Concurrent cisplatin	**40 mg/m² IV weekly × 5–6**; withhold if creatinine clearance below about **50 mL/min**
Brachytherapy (HDR)	e.g. **24–28 Gy in 3–4 fractions**
Overall treatment time	**≤56 days**
Hb during radiotherapy	Keep **≥10 g/dL**; transfuse if **<7 g/dL**
Shock index	**≥0.9** abnormal; **≥1.4** urgent intervention
Heavy tumour bleeding	**Tranexamic acid 1 g IV 8-hourly**, vaginal pack + catheter (24–48 h), haemostatic RT
Oral morphine (opioid-naive)	**2.5–5 mg every 4 hours** + same dose for breakthrough; always a laxative
Offensive discharge	**Metronidazole 400 mg PO 8-hourly** (or topically in packs)
Bone pain RT	Single **8 Gy** fraction
Radical hysterectomy	Removes upper third (about **2 cm**) of vagina
Late presentation in Uganda	About **80%** of women at UCI have stage IIB or worse; **12%** completed full course in one UCI study
5-year survival (with treatment)	Stage I **80–95%**, II **60–75%**, III **30–45%**, IVA **15–20%**, IVB **<10%**
Follow-up	Every **3–6 months for 2 years**, then 6–12-monthly to 5 years

Exam traps

- **"VIA was negative so she does not have cancer."** Wrong: a **symptomatic** woman is **diagnosed, not screened**; persistent PCB needs colposcopy and biopsy (adenocarcinoma can hide in the canal).
- **Staging a 2 cm tumour with hydronephrosis as stage I or II.** Wrong: hydronephrosis or a non-functioning kidney = **IIIB**, whatever the tumour size (unless another cause).
- **Calling bullous oedema of the bladder IVA.** Wrong: IVA needs **biopsy-proven mucosal** invasion.
- **Ablating a suspicious lesion because it is "VIA positive".** Wrong: a growth or ulcer is **suspicious for cancer** → biopsy and refer; ablation destroys the diagnosis.
- **Biopsying the necrotic centre.** Wrong: take the punch biopsy from the **edge** of the growth.
- **Forgetting the rectovaginal examination.** Wrong: it is the only bedside way to assess the **parametria** (IIB vs IIIB).
- **Starting cisplatin without checking the kidneys.** Wrong: check **creatinine and renal ultrasound** first; drain obstructed kidneys before curative chemoradiation.
- **Choosing surgery for IB3 or IIB.** Wrong: locally advanced disease (IB3–IVA) gets **concurrent chemoradiation**; avoid surgery + radiation "double treatment".

Questions seniors ask

Viva questions

1. *How does HPV cause cancer?* Persistent high-risk HPV integrates, E2 is lost, E6 degrades p53 and E7 inactivates Rb → uncontrolled proliferation and genomic instability.
2. *What are the WHO 90-70-90 targets?* 90% of girls fully vaccinated by 15; 70% of women screened with a high-performance test by 35 and again by 45; 90% of women with pre-cancer treated and 90% with invasive cancer managed.
3. *How is VIA performed and read?* 3–5% acetic acid, 1 minute; dense acetowhite lesion at the SCJ = positive; growth/ulcer = suspicious for cancer.

4. *Eligibility criteria for ablation?* Entire lesion and SCJ visible, <75% of the cervix, no extension into the canal/vagina, no suspicion of cancer, not pregnant.
5. *Stage a woman with a 3 cm tumour and hydronephrosis.* IIIB (hydronephrosis upstages to IIIB regardless of size, unless another cause).
6. *Why is cisplatin given with radiotherapy?* Radiosensitiser; improves survival; weekly 40 mg/m².
7. *What is removed in a radical hysterectomy?* Uterus, cervix, parametria, upper vagina, pelvic nodes (ovaries can be conserved).
8. *Commonest cause of death?* Renal failure from bilateral ureteric obstruction.
9. *Why is anaemia corrected during radiotherapy?* Hypoxic tumour cells are radioresistant; keep Hb ≥10 g/dL.
10. *How is cervical cancer in pregnancy delivered?* Caesarean section (avoid haemorrhage and tumour seeding of the episiotomy).

On the ward round

- **How do you assess the parametrium?** Rectovaginal examination (ideally EUA): thickened with a gap to the side wall = IIB; reaching the side wall = IIIB. MRI where available.
- **What is the minimum staging workup you can do before referral?** Biopsy, Hb, urea/creatinine, renal ultrasound, CXR, HIV status (± CD4/VL), and a careful speculum, bimanual and rectovaginal examination.
- **This woman has a 3 cm tumour and left hydronephrosis: what stage and what treatment?** IIIB; concurrent chemoradiation if renal function allows (nephrostomy if needed), otherwise radiotherapy alone or palliation.
- **Why give cisplatin with radiotherapy, and what do you monitor?** Radiosensitiser, improves survival; weekly creatinine and electrolytes (Mg, K), FBC, hearing, emesis.
- **Why must treatment finish within 56 days?** Tumour cells repopulate during gaps; each extra day reduces local control.
- **How do you stop heavy bleeding from a cervical tumour?** Resuscitate, TXA 1 g IV, tight vaginal pack + urinary catheter, transfuse, urgent haemostatic radiotherapy; embolisation/ligation where available.
- **Why is renal failure the commonest cause of death?** Parametrial and side-wall spread compresses both ureters → obstructive uropathy.
- **What would you tell a woman with stage IVB disease?** Honest explanation of the diagnosis; aims of care (comfort, bleeding, odour and pain control); oral morphine is safe and free; link to palliative care and hospice the same day; involve family if she wishes.

Mnemonics

- **"E6 kills p53, E7 robs Rb"**: **6** → p**53**, **7** → **R**b ("E7 Robs").
- **FIGO sizes, "2, 4"**: IB1 <2, IB2 2–4, IB3 ≥4; IIA1 <4, IIA2 ≥4.
- **Stage III = "3 Ls"**: **L**ower third of vagina (IIIA), **L**ateral pelvic wall/hydronephrosis, the **L**oin (IIIB), **L**ymph nodes (IIIC).
- **Stage IV = "neighbours then far away"**: IVA bladder/rectum mucosa; IVB distant.
- **90-70-90**: "**V**accinate – **S**creen – **T**reat" (VST).
- **Ablation eligibility, "SEEN"**: **S**CJ visible, **E**ctocervical, **E**xtent <75%, **N**o suspicion of cancer.
- **Cryotherapy, "3-5-3"**: freeze 3 minutes, thaw 5, freeze 3.
- **Chemoradiation, "45–50 / 40 / 56"**: EBRT 45–50 Gy, cisplatin 40 mg/m² weekly, finish within 56 days.

OSCE tips

> EXAM FAVOURITE
> **OSCE: counselling for screening**: explain what HPV is (common, sexually transmitted, usually clears), why screening (find pre-cancer early, simple treatment), how (self-sample or speculum), what a positive result means (not cancer, but needs VIA/treatment), treatment (thermal ablation: quick, little pain, same day), after-care (discharge, abstain 4 weeks), follow-up (1 year), and check understanding. Avoid blame: HPV is not a sign of unfaithfulness.

- **VIA station**: speculum, identify the SCJ, 3–5% acetic acid, wait 1 minute, report negative / positive / suspicious for cancer, state ablation eligibility.
- **Staging station**: describe size, vaginal extent, each parametrium separately (rectovaginal), rectal mucosa, nodes and kidneys, then assign the FIGO stage and say what changes it.
- **Breaking bad news / palliative station**: honest explanation, aims of care, morphine is safe and free, same-day palliative care link, involve family if she wishes.

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SELF-TEST

Flashcards

25 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Which HPV types cause about 70% of cervical cancers?	**HPV 16 and 18** (high-risk). HPV 6 and 11 are low-risk and cause genital warts.
2	What do HPV E6 and E7 do?	**E6** binds E6AP and targets **p53** for degradation (no apoptosis, genomic instability, telomerase activation). **E7** inactivates **Rb**, releasing E2F → uncontrolled S-phase entry. Loss of viral **E2** on integration drives their over-expression.
3	Differentiate LSIL and HSIL.	**LSIL** = CIN 1 / HPV effect (lower third of epithelium; usually regresses). **HSIL** = CIN 2 and CIN 3 (two-thirds to full thickness; true pre-cancer — treat).
4	WHO 2021 screening for the general population of women.	**HPV DNA** primary test from age **30**, every **5–10 years** (VIA/cytology every 3 years where HPV is not available); screen-and-treat or screen-triage-treat; re-test **12 months** after treatment.
5	WHO 2021 screening for women living with HIV.	**HPV DNA** from age **25**, every **3–5 years**, using **screen, triage and treat**; re-test at **12 months** after treatment.
6	State the WHO 90-70-90 targets for 2030.	**90%** of girls fully vaccinated against HPV by 15; **70%** of women screened with a high-performance test by **35 and again by 45**; **90%** of women with pre-cancer treated and with invasive cancer managed. Elimination = incidence **<4 per 100,000** women.
7	How is VIA performed and interpreted?	Apply **3–5% acetic acid**, wait **1 minute**. **Negative**: no acetowhite. **Positive**: dense, well-defined acetowhite lesion touching/adjacent to the SCJ. **Suspicious for cancer**: cauliflower growth or ulcer → biopsy.
8	Criteria for a lesion to be eligible for ablation.	Whole lesion and **SCJ fully visible** (TZ type 1/2); lesion **<75%** of the ectocervix; **no extension into the canal** or vagina; **no suspicion of invasive or glandular disease**; not pregnant. Otherwise → LEEP or biopsy.
9	Thermal ablation and cryotherapy technique.	**Thermal ablation**: probe at **≥100 °C** for **20–30 s** per application, overlapping to cover the TZ. **Cryotherapy**: CO_2/N_2O **double freeze — 3 min freeze, 5 min thaw, 3 min freeze**.
10	HPV vaccination in Uganda.	Target **10-year-old girls**; Uganda moved to a **single dose** in 2025 (WHO 2022 allows 1 or 2 doses for 9–20-year-olds). **Immunocompromised/HIV: at least 2 doses, ideally 3.** 0.5 mL IM deltoid.
11	FIGO 2018: define IA1, IA2, IB1, IB2, IB3.	**IA1** invasion <3 mm deep; **IA2** ≥3 to <5 mm; **IB1** ≥5 mm depth and <2 cm; **IB2** 2 to <4 cm; **IB3** ≥4 cm — all confined to the cervix.
12	FIGO 2018 stage for: parametrial involvement; pelvic wall or hydronephrosis; pelvic nodes; para-aortic nodes.	**IIB**; **IIIB**; **IIIC1**; **IIIC2**.
13	FIGO stage IVA vs IVB.	**IVA**: biopsy-proven invasion of **bladder or rectal mucosa** (bullous oedema alone is not enough). **IVB**: **distant** metastasis (lung, liver, bone, distant nodes).
14	Treatment of stage IB1–IB2 and IIA1 cervical cancer.	**Radical hysterectomy + bilateral pelvic lymphadenectomy** (ovaries can be conserved) **or** primary chemoradiation. Adjuvant chemoradiation if nodes, parametria or margins positive (Peters).

#	Question	Answer
15	Standard chemoradiation for locally advanced cervical cancer (IB3–IVA).	**EBRT 45–50 Gy in 25 fractions** + **cisplatin 40 mg/m² IV weekly × 5–6** + **brachytherapy** (e.g. 24–28 Gy in 3–4 HDR fractions); total treatment time **≤56 days**.
16	What is removed in a radical (Wertheim) hysterectomy?	Uterus, cervix, **parametria**, **upper vagina** (about 2 cm) and **pelvic lymph nodes**; ovaries may be conserved in young women with squamous cancer.
17	Commonest cause of death in cervical cancer?	**Renal failure (uraemia)** from bilateral **ureteric obstruction** by parametrial/pelvic wall disease.
18	Why keep Hb ≥10 g/dL during radiotherapy?	Anaemia causes **tumour hypoxia**; hypoxic cells are **radioresistant** (radiation kills via oxygen-dependent free radicals), so local control falls.
19	First steps for heavy bleeding from a cervical tumour.	ABC, 2 large-bore IV lines, group and cross-match, fluids/transfusion; **tranexamic acid 1 g IV 8-hourly**; **tight vaginal pack** + **urinary catheter**; urgent **haemostatic radiotherapy**; embolisation/internal iliac ligation if available.
20	Starting oral morphine for cancer pain (opioid-naive, normal renal function).	Immediate-release oral morphine **2.5–5 mg every 4 hours** + same dose for breakthrough; titrate daily; **always co-prescribe a laxative**; reduce dose/frequency in renal failure. Free in Ugandan public facilities.
21	How do you assess the parametria clinically, and what distinguishes IIB from IIIB?	**Rectovaginal examination** (ideally at **EUA**): finger in the vagina and rectum, feel between the cervix and the pelvic side wall on **each side separately**. **Thickened but a gap remains** to the side wall = **IIB**; **thickening reaches/fixed to the side wall** (frozen pelvis) = **IIIB**.
22	List the minimum staging and fitness workup you can complete at a general hospital before referring a woman with cervical cancer to UCI.	**Punch biopsy** (histology), **FBC/Hb**, **urea, creatinine and electrolytes**, **renal ultrasound** (hydronephrosis), **chest X-ray**, **HIV status** (± CD4, viral load), plus a documented speculum, bimanual and rectovaginal examination with tumour size and vaginal/parametrial extent.
23	A 4 cm cervical tumour with free parametria, but renal ultrasound shows right hydronephrosis. FIGO 2018 stage?	**IIIB**. Hydronephrosis or a non-functioning kidney upstages to IIIB **regardless of tumour size or parametrial findings** (unless known to be from another cause).
24	Cystoscopy shows bullous oedema of the bladder mucosa but no tumour. Is this stage IVA?	**No.** IVA requires **biopsy-proven** tumour in the **bladder or rectal mucosa**. Bullous oedema alone does not upstage. Stage on the other findings.
25	A 55-year-old, 50 kg, creatinine 150 µmol/L, is referred for chemoradiation. Can she have cisplatin?	Cockcroft–Gault (women) ≈ (140 − 55) × 50 × 1.04 ÷ 150 ≈ **29 mL/min**: below about **50 mL/min**, so **no cisplatin** as things stand. Look for **obstructive uropathy** (IIIB) → **nephrostomy**, recheck renal function; otherwise RT alone/carboplatin or palliation per the oncologist (unit cut-offs vary).

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 34-year-old HIV-negative woman attends a HC IV for screening. VIA shows a dense, well-defined acetowhite lesion at 3 o'clock touching the SCJ, covering about 30% of the ectocervix; the entire SCJ is visible and there is no growth or ulcer. She is not pregnant. What is the MOST appropriate management?

- A Refer for punch biopsy and await histology before any treatment
- B Offer thermal ablation at the same visit (screen-and-treat)
- C Perform a total abdominal hysterectomy
- D Repeat VIA in 3 years
- E Refer to the Uganda Cancer Institute for chemoradiation

2. A 29-year-old woman living with HIV (on TLD, viral load suppressed) asks when she should start cervical screening and how often. According to WHO 2021, what is the BEST answer?

- A Start at 30, HPV test every 10 years
- B Start at 21, cytology every year
- C Start at 35, VIA once in a lifetime
- D Start now (from 25), HPV DNA every 3–5 years with triage of positives
- E Screening is not needed if the viral load is suppressed

3. During VIA in a 41-year-old woman, you see an acetowhite lesion that occupies about 80% of the ectocervix and disappears into the endocervical canal; you cannot see its upper limit. There is no growth or ulcer. What is the MOST appropriate management?

- A Refer for LEEP (loop excision) at a hospital with the service
- B Treat with cryotherapy today
- C Treat with thermal ablation today
- D Reassure and repeat in 3 years
- E Start chemoradiation

4. A 50-year-old P8 has a 3 cm cervical tumour confined to the cervix on examination with free parametria. Renal ultrasound shows moderate right hydronephrosis with no stone or other cause. Chest X-ray is clear. Biopsy shows squamous cell carcinoma. What is the FIGO 2018 stage?

- A IB2
- B IIA1
- C IIB
- D IVA
- E IIIB

5. A 38-year-old P3 has a 2.5 cm squamous cell carcinoma confined to the cervix (FIGO IB2) with no nodal disease on imaging. She is fit, HIV-negative and has completed her family. Which is the MOST appropriate primary treatment?

- A Cone biopsy alone
- B Simple (extrafascial) hysterectomy
- C Radical hysterectomy with bilateral pelvic lymphadenectomy, conserving the ovaries
- D Palliative chemotherapy
- E Thermal ablation

6. A 45-year-old woman with FIGO IIB cervical squamous carcinoma is booked for chemoradiation at the Uganda Cancer Institute. Her Hb is 7.8 g/dL and creatinine 85 µmol/L. Which statement is MOST correct? Advanced

A Cisplatin is contraindicated because she has stage II disease
B Transfuse to keep Hb about 10 g/dL or more, because hypoxic tumour cells are radioresistant
C Radical hysterectomy is preferred over chemoradiation for IIB
D Brachytherapy can be omitted if EBRT is given
E Treatment can safely be spread over 12 weeks to reduce side effects

7. A 55-year-old woman with known stage IIIB cervical cancer is brought to a general hospital with heavy fresh vaginal bleeding. BP 86/50 mmHg, pulse 124/min, Hb 5.9 g/dL. After calling for help and inserting two large-bore IV cannulae, what is the MOST appropriate next step?

A Emergency radical hysterectomy
B Cryotherapy to the tumour
C Oral ferrous sulphate and discharge
D Resuscitate with fluids and blood, give tranexamic acid 1 g IV, pack the vagina tightly and insert a urinary catheter, then refer for haemostatic radiotherapy
E Start weekly cisplatin immediately

8. A 60-year-old woman with advanced cervical cancer has constant severe pelvic pain not relieved by paracetamol 1 g four times a day and ibuprofen. Her creatinine is normal. What is the MOST appropriate next step in pain management?

A Start immediate-release oral morphine 5 mg every 4 hours with breakthrough doses and a regular laxative
B Increase paracetamol to 2 g four times a day
C Give pethidine 100 mg IM as needed
D Avoid opioids because of the risk of addiction
E Refer for surgical debulking

9. A mother asks why her 10-year-old daughter is being offered HPV vaccine at school when she is 'too young for sex'. Which explanation is MOST accurate?

A The vaccine treats existing HPV infection, so it is given early to clear it
B It is given at 10 because older girls cannot receive it
C It removes the need for cervical screening later in life
D The vaccine is only effective in children under 12 because of the thymus
E The vaccine prevents infection and works best before any exposure to HPV, which is sexually transmitted; young girls also produce a very strong immune response

10. A 48-year-old woman with FIGO IIIB cervical cancer presents with vomiting, hiccups, reduced urine output and pruritus. Creatinine is 780 µmol/L and ultrasound shows bilateral hydronephrosis. She is weak and bed-bound (ECOG 4). After discussion she says she does not want invasive procedures. What is the MOST appropriate approach? Advanced

A Start cisplatin-based chemoradiation immediately
B Insist on bilateral nephrostomy tubes
C Focus on comfort care: symptom control (antiemetic, analgesia), family support and hospice/home-based palliative care
D Perform an emergency radical hysterectomy
E Start haemodialysis three times weekly indefinitely

11. A 31-year-old woman, 20 weeks pregnant, is found at antenatal clinic to have a VIA-positive acetowhite lesion with no features of invasion. What is the MOST appropriate management? Advanced

A Perform thermal ablation immediately

B Colposcopy (and biopsy only if invasion is suspected), and defer treatment of pre-cancer until at least 6 weeks postpartum

C Terminate the pregnancy and perform a hysterectomy

D Perform LEEP now to avoid progression

E Tell her that screening results are invalid in pregnancy and discharge her

12. After radical hysterectomy and pelvic lymphadenectomy for stage IB2 cervical cancer, histology shows metastatic carcinoma in two left pelvic lymph nodes with clear margins and parametria. What is the revised stage and MOST appropriate adjuvant management? Advanced

A Stage IB2; no further treatment

B Stage IIB; vaginal brachytherapy alone

C Stage IVB; palliative care only

D Stage IIIC1p; adjuvant concurrent chemoradiation

E Stage IIA1; repeat surgery

13. Which molecular event BEST explains why HPV integration into the host genome promotes malignant transformation?

A Loss of the viral E2 gene, removing repression of E6 and E7, which then inactivate p53 and Rb

B Activation of the viral L1 protein, which causes cell lysis

C Direct insertion of HPV into the BRCA1 gene

D Increased production of viral capsid proteins in basal cells

E Viral induction of oestrogen receptors

14. A 42-year-old woman presents to a HC IV with post-coital bleeding. VIA is 'negative', but you notice the cervix is barrel-shaped, hard and enlarged, and it bleeds on touch. What is the MOST appropriate next step? Advanced

A Reassure her because VIA is negative

B Treat for cervicitis and repeat VIA in 3 years

C Offer thermal ablation

D Give tranexamic acid and review in 6 months

E Refer urgently for colposcopy, biopsy (including endocervical sampling) and staging, as endocervical adenocarcinoma is suspected

15. A 36-year-old woman is treated with cryotherapy for a VIA-positive lesion. Which after-care advice is MOST appropriate?

A She should return in 5 years for re-screening

B Heavy bleeding for 2 weeks is expected and does not need review

C Expect a watery discharge for 2–4 weeks, avoid intercourse (or use condoms) and tampons for about 4 weeks, return if fever, offensive discharge or heavy bleeding, and re-screen at 12 months

D She must have a hysterectomy after 6 months

E She should avoid bathing for a month

Answers & explanations

1. B · Offer thermal ablation at the same visit (screen-and-treat)

B is correct. A VIA-positive lesion that is fully visible, <75% of the cervix, not entering the canal and not suspicious for cancer is **eligible for ablation**, ideally the **same day** (screen-and-treat) to avoid loss to follow-up. **A** is wrong: biopsy first adds visits and is reserved for suspected cancer or non-eligible lesions. **C** is wrong: hysterectomy is gross over-treatment of pre-cancer. **D** is wrong: a positive test needs treatment, not routine re-screening. **E** is wrong: this is not invasive cancer.

2. D · Start now (from 25), HPV DNA every 3–5 years with triage of positives

D is correct. WHO recommends that women living with HIV start screening at **25** with **HPV DNA** every **3–5 years**, using a **screen-triage-treat** approach. **A** is the general-population recommendation (30, every 5–10 years). **B** is a North American cytology schedule, not WHO. **C** is inadequate. **E** is wrong: ART reduces but does not remove the elevated risk of persistent HPV and CIN.

3. A · Refer for LEEP (loop excision) at a hospital with the service

A is correct. The lesion is **not eligible for ablation** — it covers >75% of the cervix and extends into the canal, so the SCJ and upper lesion cannot be seen. **Excision (LEEP)** removes it and provides histology. **B** and **C** are wrong: ablation cannot reliably reach a lesion in the canal and would under-treat possible occult invasion. **D** is wrong: a positive screen must be managed. **E** is wrong: there is no histological evidence of invasive cancer.

4. E · IIIB

E is correct. Hydronephrosis or a non-functioning kidney (not due to another cause) places the tumour in **stage IIIB**, regardless of its size or parametrial findings on examination — it implies ureteric obstruction by parametrial/pelvic wall disease. **A** would be correct only without hydronephrosis. **B** requires upper vaginal involvement. **C** requires parametrial involvement without hydronephrosis. **D** requires bladder/rectal mucosal invasion.

5. C · Radical hysterectomy with bilateral pelvic lymphadenectomy, conserving the ovaries

C is correct. For IB1–IB2 (and IIA1), **radical hysterectomy + pelvic lymphadenectomy** (or primary chemoradiation) is standard; surgery in a young woman preserves ovarian function and vaginal elasticity. **A** and **B** are adequate only for IA1 disease. **D** is wrong: she has curable disease. **E** is wrong: ablation is for pre-cancer only.

6. B · Transfuse to keep Hb about 10 g/dL or more, because hypoxic tumour cells are radioresistant

B is correct. Anaemia causes tumour **hypoxia**, and radiation kills cells mainly through oxygen-dependent free radicals, so correcting Hb improves local control. **A** is wrong: weekly cisplatin 40 mg/m² is standard for IIB with normal renal function. **C** is wrong: parametrial involvement (IIB) is treated with chemoradiation, not surgery. **D** is wrong: brachytherapy is essential — omitting it markedly reduces survival. **E** is wrong: total treatment time should be **≤56 days**; prolongation allows tumour repopulation.

7. D · Resuscitate with fluids and blood, give tranexamic acid 1 g IV, pack the vagina tightly and insert a urinary catheter, then refer for haemostatic radiotherapy

D is correct. Tumour haemorrhage is controlled by **resuscitation, tranexamic acid, tight vaginal packing (with a urinary catheter because the pack compresses the urethra)** and then **haemostatic radiotherapy**. **A** is wrong: radical surgery is impossible and dangerous in stage IIIB. **B** is wrong: cryotherapy is for pre-cancer and would not stop bleeding from a large tumour. **C** is wrong: she is in shock. **E** is wrong: chemotherapy has no immediate haemostatic effect and she is unstable.

8. A · Start immediate-release oral morphine 5 mg every 4 hours with breakthrough doses and a regular laxative

A is correct. Following the WHO analgesic ladder, severe pain uncontrolled by non-opioids needs a **strong opioid**: oral morphine 2.5–5 mg 4-hourly with the same dose for breakthrough, titrated daily, **always with a laxative**. Oral morphine is free in Uganda. **B** is wrong: exceeding 4 g/day of paracetamol causes hepatotoxicity. **C** is wrong: pethidine is short-acting with a toxic metabolite and IM injections are unsuitable for chronic pain. **D** is wrong: addiction is not a relevant concern in cancer pain; under-treatment is the real problem. **E** is wrong: debulking has no role in advanced cervical cancer.

9. E · The vaccine prevents infection and works best before any exposure to HPV, which is sexually transmitted; young girls also produce a very strong immune response

E is correct. HPV vaccines are **prophylactic** — they induce neutralising antibodies that prevent infection, so they work best **before sexual debut**, and 9–14-year-olds mount very high antibody responses (which is why fewer doses suffice). **A** is wrong: the vaccine does not treat established infection. **B** is wrong: older girls and women can be vaccinated (catch-up), although benefit falls with prior exposure. **C** is wrong: screening is still needed. **D** is wrong: the thymus explanation is invented.

10. C · Focus on comfort care: symptom control (antiemetic, analgesia), family support and hospice/home-based palliative care

C is correct. She has terminal obstructive uropathy with very poor performance status, and she has declined intervention; **comfort-focused palliative care** respecting her wishes is appropriate. Uraemia can be managed as a relatively peaceful terminal event with good symptom control. **A** is wrong: cisplatin is contraindicated in renal failure and she is too frail. **B** is wrong: nephrostomy is an option for candidates for active treatment, but it cannot be imposed against a competent patient's wishes. **D** is wrong: surgery has no role. **E** is wrong: dialysis is rarely available and not appropriate here.

11. B · Colposcopy (and biopsy only if invasion is suspected), and defer treatment of pre-cancer until at least 6 weeks postpartum

B is correct. Pre-cancer progresses slowly; in pregnancy the priority is to **exclude invasion** (colposcopy ± biopsy if suspicious) and to **defer treatment until ≥6 weeks postpartum**. **A** and **D** are wrong: ablation/excision in pregnancy risks bleeding, miscarriage and preterm birth without benefit. **C** is grossly inappropriate for pre-cancer. **E** is wrong: a positive test must be followed up.

12. D · Stage IIIC1p; adjuvant concurrent chemoradiation

D is correct. Pathologically positive pelvic nodes upstage to **IIIC1p** under FIGO 2018, and positive nodes are a **Peters high-risk** criterion requiring **adjuvant chemoradiation** (pelvic EBRT with concurrent cisplatin). **A** ignores the node status. **B** and **E** mis-stage the disease (parametria are clear). **C** is wrong: pelvic nodes are not distant metastases and the disease is still potentially curable.

13. A · Loss of the viral E2 gene, removing repression of E6 and E7, which then inactivate p53 and Rb

A is correct. Integration typically disrupts **E2**, the viral repressor of E6/E7; over-expressed **E6 degrades p53** and **E7 inactivates Rb**, driving proliferation and genomic instability. **B** is wrong: L1 is the capsid protein (the vaccine antigen) and does not lyse cells. **C** is wrong: HPV integration is not targeted to BRCA1. **D** is wrong: capsid production occurs in superficial differentiated cells during productive infection, not transformation. **E** is invented.

14. E · Refer urgently for colposcopy, biopsy (including endocervical sampling) and staging, as endocervical adenocarcinoma is suspected

E is correct. A **barrel-shaped, hard, bleeding cervix** suggests an **endophytic/endocervical tumour** (often adenocarcinoma) that VIA cannot see; symptomatic women need **diagnostic assessment and biopsy**, not screening reassurance. **A**, **B** and **D** all falsely reassure. **C** is wrong: ablation is contraindicated when cancer is suspected.

15. C · Expect a watery discharge for 2–4 weeks, avoid intercourse (or use condoms) and tampons for about 4 weeks, return if fever, offensive discharge or heavy bleeding, and re-screen at 12 months

C is correct. Watery discharge is expected after ablation as the necrotic tissue sloughs; abstinence/condoms for about 4 weeks allows healing (and reduces HIV transmission risk); warning signs of infection or haemorrhage need review; WHO recommends **re-testing at 12 months** after treatment. **A** is wrong: post-treatment follow-up is at 12 months. **B** is wrong: heavy bleeding is abnormal. **D** is wrong: ablation is definitive treatment for pre-cancer. **E** is wrong: bathing is fine.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Screen-and-treat at an HIV clinic

Apio, a 32-year-old P3 living with HIV (on TLD for 6 years, viral load undetectable), attends the ART clinic at a HC IV in Soroti. She has never been screened for cervical cancer. The clinic has a GeneXpert machine for HPV testing and a thermal ablation device.

1. Why is Apio a priority for screening and what does WHO recommend for her?

> Women living with HIV have about a **6-fold** higher risk of cervical cancer (reduced HPV clearance, faster progression). WHO 2021: **HPV DNA** screening from **25**, every **3–5 years**, using **screen-triage-treat**, re-test at **12 months** after treatment. She is 32 and never screened — screen today.

2. How would you counsel her before the test?

> HPV is a very common sexually transmitted virus that usually clears; persistent infection can cause pre-cancer, which is easily treated; explain **self-sampling** (or clinician sample) for HPV; a positive result does **not** mean cancer; if positive she will have a VIA look and, if suitable, **same-day thermal ablation**; address fears (pain, infidelity) and consent.

3. Her HPV test is positive. VIA triage shows an acetowhite lesion at 6–8 o'clock covering about 40% of the cervix, SCJ fully visible, no growth. She is not pregnant. What do you do?

> The lesion is **eligible for ablation** (fully visible, <75%, not in canal, no suspicion of cancer). Perform **thermal ablation** today: probe at **≥100 °C for 20–30 seconds** per overlapping application to cover the whole TZ. No anaesthesia is usually needed; offer paracetamol.

4. What after-care and follow-up does she need?

> Watery discharge for 2–4 weeks; **no intercourse or tampons for 4 weeks** (condoms if unavoidable — also reduces HIV transmission during healing); return for fever, offensive discharge, heavy bleeding or severe pain. **Re-test (HPV) at 12 months**, then every 3–5 years if negative. Continue ART adherence.

5. What if the lesion had entered the canal, or a friable growth had been seen?

> Canal involvement or >75% coverage → **refer for LEEP** (excision gives histology). A friable growth/ulcer → **suspicious for cancer**: do **not** ablate; take a **punch biopsy** from the edge, check creatinine/renal ultrasound, and refer urgently to a gynae-oncology service.

Take-away: Screen–triage–treat in one visit: HPV test, VIA to triage, same-day ablation if eligible, LEEP if not, biopsy if cancer is suspected — and re-test women living with HIV at 12 months.

Case 2: Offensive discharge and back pain

Mrs Nabirye, a 52-year-old P7 from Jinja, comes to Mengo gynaecology ward with 8 months of post-coital bleeding progressing to continuous offensive blood-stained discharge, low back pain radiating down the left leg and weight loss. She has never been screened. HIV status unknown.

1. What else do you ask in the history?

> Urine or faeces per vagina (fistula); urine output, vomiting, hiccups, itching (uraemia); leg swelling; bleeding volume and dizziness (anaemia); bone pain, cough; menstrual/menopausal status; HIV testing history; family support, finances, distance to Kampala (ability to attend UCI); her understanding and wishes.

2. On examination she is pale (Hb 7.0 g/dL), with left leg oedema. Speculum: a 6 cm fungating, necrotic growth replacing the cervix, extending to the upper vagina. Rectovaginal: left parametrium thickened and fixed to the pelvic side wall; right parametrium thickened but not reaching the wall. What is your clinical stage and what investigations complete the work-up?

> Clinically at least **FIGO IIIB** (left pelvic side-wall involvement). Investigations: **punch biopsy from the edge** of the growth; FBC, group and cross-match; **urea, creatinine, electrolytes**; **renal/abdominal ultrasound** (hydronephrosis, liver metastases); **chest X-ray**; **HIV test** (CD4/viral load if positive); CT/MRI if accessible; consider EUA with cystoscopy if bladder invasion suspected.

3. Biopsy: squamous cell carcinoma. Creatinine 95 µmol/L; mild left hydronephrosis; CXR clear. HIV-negative. What is the definitive treatment plan?

> Stage **IIIB → concurrent chemoradiation** at UCI: **EBRT 45–50 Gy in 25 fractions** + **weekly cisplatin 40 mg/m²** (renal function adequate) + **HDR brachytherapy** (e.g. 24–28 Gy in 3–4 fractions), all within **56 days**. Transfuse to **Hb ≥10 g/dL**. Nutritional support. Prompt referral with biopsy report and staging results.

4. What symptom control do you start today while she waits for UCI?

> Pain: paracetamol 1 g 6-hourly + **oral morphine 2.5–5 mg 4-hourly** titrated, with a laxative; amitriptyline 12.5–25 mg nocte for sciatic neuropathic pain. Odour: **metronidazole 400 mg PO 8-hourly** ± topical. Bleeding: tranexamic acid 1 g PO 8-hourly. Leg: exclude DVT (Doppler) — elevation. **Link to palliative care** and counsel her and family.

5. Six months later she returns with anuria, vomiting and creatinine 900 µmol/L; she never started radiotherapy because of cost and distance. How do you approach this?

> Bilateral obstructive uropathy — the **commonest cause of death**. Hold an honest family meeting: options are **nephrostomy** (only sensible if active cancer treatment is realistic) versus **comfort care**. Respect her wishes; control nausea (haloperidol), pain (reduce morphine dose/frequency — accumulation in renal failure), itching; hospice/home-based care; spiritual support.

> **Take-away:** Stage clinically with a rectovaginal examination plus kidney ultrasound, biopsy the edge, refer for chemoradiation within an 8-week treatment window, and start palliative care on the day of diagnosis — most Ugandan women need both.

Case 3: Early disease in a young mother

Ms Kemigisha, a 33-year-old P2 teacher from Mbarara, had a VIA-positive lesion that was referred for LEEP because it entered the canal. The LEEP histology shows invasive squamous cell carcinoma with 4 mm depth of invasion, no lymphovascular space invasion, and positive endocervical margin.

1. What stage is this likely to be and what additional information do you need?

> Depth 4 mm with no visible lesion = at least **FIGO IA2** (≥3 to <5 mm) — but the margin is **positive**, so the true extent is unknown (it could be IB1). Need: examination/colposcopy, **MRI pelvis** (tumour size, parametria, nodes) or at least clinical examination + ultrasound, CXR, renal function, HIV status, and her **fertility wishes**.

2. MRI shows no residual visible tumour and no enlarged nodes. She wants more children. What are the options?

> For IA2 wanting fertility: **repeat cone to achieve clear margins + pelvic lymphadenectomy**, or **radical trachelectomy + pelvic lymphadenectomy** (removes cervix and parametria, keeps the uterus; a cerclage is placed). If family is complete: **modified radical hysterectomy + pelvic lymphadenectomy** (ovaries conserved). Discuss at the gynae-oncology MDT.

3. She chooses modified radical hysterectomy with pelvic lymphadenectomy. What complications will you counsel her about?

> Bleeding, infection, **ureteric injury/ureterovaginal fistula**, bladder injury and **voiding dysfunction** (autonomic nerve damage — may need a catheter for days to weeks), **lymphocyst**, **lower-limb lymphoedema**, DVT, shortened vagina/dyspareunia; loss of fertility; possible need for adjuvant (chemo)radiation depending on histology.

4. Histology: 1.2 cm residual tumour, clear margins and parametria, 0/18 nodes positive. What is the follow-up?

> No adjuvant treatment needed (no Peters high-risk factors; assess Sedlis intermediate-risk factors — small tumour, likely none). Follow-up every **3–6 months for 2 years**, then 6–12-monthly to 5 years: symptoms, speculum/vault examination, pelvic exam. Report new bleeding, pain, leg swelling promptly. Ovaries conserved — no menopause.

Take-away: Staging depends on histology depth and margins as well as imaging; in early disease, fertility wishes decide between cone/trachelectomy and radical hysterectomy, and node status decides adjuvant therapy.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.